

Rigorous Proofs for “Genetic Testing: Assessing Severity and Implications for Health Insurance”

Revised Proofs

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1 Setup and Notation

Before presenting the proofs, we establish formal assumptions and notation that the paper should state explicitly.

Assumption 1 (Preferences). $u : \mathbb{R} \rightarrow \mathbb{R}$ is C^3 , with $u' > 0$, $u'' < 0$, and $u''' > 0$ (prudence in the sense of Kimball 1990). u is defined on a domain wide enough that all wealth realizations in what follows lie in its interior.

Assumption 2 (Parameters). $0 < L < H < w$, $\mu \in (0, 1)$, $\theta_H, \theta_L \in (0, 1)$ with $\theta_H + \theta_L = 1$. Let $A := \theta_H H + \theta_L L \in (L, H)$.

Assumption 3 (Insurance Market). Insurers are risk-neutral, competitive, and earn zero expected profit on each contract sold. A contract is a pair $C = (q, d)$ where q is the premium rate and d the indemnity; the premium is $\pi = qd$. With the principle of indemnity, the payout in state i is $\min\{i, d\}$; without it, the payout is d .

State-Contingent Wealth

With contract (q, d) :

- No-loss wealth: $W_0(q, d) := w - qd$
- Loss wealth without indemnity principle: $W_1^{np}(q, d, i) := w - qd - i + d$
- Loss wealth with indemnity principle: $W_1^p(q, d, i) := w - qd - i + \min\{i, d\}$

2 Proposition 1: Uninformed Optimal Coverage Without Principle of Indemnity

Proposition 1 (Existence, uniqueness, and characterization of $d_{U,np}^*$). Suppose insurance is actuarially fair, $q = \mu$, and the principle of indemnity is absent. The uninformed individual’s problem

$$V_{U,np} = \max_{d \geq 0} \left\{ \sum_{i \in \{L, H\}} \theta_i \mu u(w - \mu d - i + d) + (1 - \mu) u(w - \mu d) \right\} \quad (1)$$

admits a unique optimum $d_{U,np}^*$, characterized by the first-order condition

$$\sum_{i \in \{H,L\}} \theta_i u'(w - \mu d - i + d) = u'(w - \mu d). \quad (2)$$

Under Assumption 1, $d_{U,np}^* > A$.

Proof. Step 1: Strict concavity of the objective. Let

$$\Phi(d) := \mu \sum_{i \in \{L,H\}} \theta_i u(w - \mu d - i + d) + (1 - \mu) u(w - \mu d).$$

Differentiating,

$$\Phi'(d) = \mu(1 - \mu) \sum_i \theta_i u'(w - \mu d - i + d) - \mu(1 - \mu) u'(w - \mu d).$$

(Here $\partial W_1^{np} / \partial d = 1 - \mu$ and $\partial W_0 / \partial d = -\mu$.) Differentiating again,

$$\Phi''(d) = \mu(1 - \mu)^2 \sum_i \theta_i u''(w - \mu d - i + d) + \mu^2(1 - \mu) u''(w - \mu d) < 0$$

since $u'' < 0$ everywhere. So Φ is strictly concave on $\mathbb{R}_+$.

Step 2: Existence of an interior optimum. Note

$$\Phi'(0) = \mu(1 - \mu) \left[\sum_i \theta_i u'(w - i) - u'(w) \right] > 0$$

because $u'(w - i) > u'(w)$ for each $i > 0$ (by $u'' < 0$). As $d \rightarrow \infty$, $W_0 \rightarrow -\infty$ and $u'(W_0) \rightarrow +\infty$. Since the no-loss term $u'(w - \mu d)$ grows unboundedly while the loss-state terms grow at the slower rate $(1 - \mu)$, eventually $\Phi'(d) < 0$ for d sufficiently large. Hence Φ' changes sign and by strict concavity has a unique zero, which is $d_{U,np}^*$.

Step 3: The FOC. Setting $\Phi'(d) = 0$ and dividing by $\mu(1 - \mu) > 0$ yields equation (2).

Step 4: $d_{U,np}^* > A$. Evaluate the FOC at $d = A$. The loss-state wealths are then

$$\begin{aligned} W_1^{np}(\mu, A, H) &= w - \mu A - H + A = w - \mu A - \theta_L(H - L) \\ W_1^{np}(\mu, A, L) &= w - \mu A - L + A = w - \mu A + \theta_H(H - L) \end{aligned}$$

Their θ_i -weighted average is exactly $w - \mu A$, which equals $W_0(\mu, A)$.

So we must determine whether

$$\theta_H u'(W_0 - \theta_L(H - L)) + \theta_L u'(W_0 + \theta_H(H - L)) \geq u'(W_0).$$

The two arguments on the left have θ -weighted mean equal to W_0 . By Assumption 1, u' is strictly convex (since $u''' > 0$). Jensen's inequality gives

$$\theta_H u'(W_0 - \theta_L(H - L)) + \theta_L u'(W_0 + \theta_H(H - L)) > u'(W_0).$$

So $\Phi'(A) > 0$, and by strict concavity the optimum $d_{U,np}^*$ lies strictly to the right of A . $\square$

Remark. The original paper's argument that "the linear combination of u' exceeds $u'(w - \pi)$ " is the same idea but presented graphically. The Jensen formulation makes the role of prudence ($u''' > 0$, equivalently u' convex) transparent.

3 Proposition 2: Testing Equilibrium Without Principle of Indemnity

We present this regime by regime. The contracts are $C_H = (\mu, H)$, $C_L = (\mu, L)$, and $C_U = (\mu, d_{U,np}^*)$, all actuarially fair.

Lemma 1. *For any uninformed contract $C_U = (\mu, d)$ with $d > 0$:*

$$\theta_H u(w - \mu H) + \theta_L u(w - \mu L) > \theta_H \mathbb{E}_\mu[u(W_1^{np}(\mu, d, H)), u(W_0(\mu, d))] + \theta_L \mathbb{E}_\mu[u(W_1^{np}(\mu, d, L)), u(W_0(\mu, d))],$$

where $\mathbb{E}_\mu[a, b] := \mu a + (1 - \mu)b$. That is, the expected utility from testing and buying full type-specific coverage strictly exceeds the expected utility from any pooled contract.

Proof. Conditional on being type i , full coverage $C_i = (\mu, i)$ delivers certain wealth $w - \mu i$, hence utility $u(w - \mu i)$. Any other contract (μ, d) with $d \neq i$ delivers a non-degenerate lottery with the same mean $w - \mu i$ (since the contract is actuarially fair: $\mathbb{E}[\text{wealth}] = w - \mu d + \mu d - \mu i = w - \mu i$). By strict concavity of u , the certain payoff strictly dominates. Taking expectations over types preserves the strict inequality. $\square$

3.1 Disclosure Duty

Game form.

1. Stage 1: Nature draws each individual's type, unobserved.
2. Stage 2: Individuals choose whether to test (test results, if produced, are publicly verifiable and must be disclosed to insurers).
3. Stage 3: Insurers post a menu of contracts, possibly conditioning on disclosed test status and result.
4. Stage 4: Individuals purchase.
5. Stage 5: Losses are realized and payouts made.

Proposition 2 (Disclosure duty equilibrium). *There exists a (subgame perfect) pure-strategy equilibrium in which*

1. *insurers offer the menu $\{C_H, C_L, C_U\}$, restricting C_i to individuals who have tested and disclosed type i and C_U to untested individuals;*
2. *every individual tests;*
3. *each type- i individual purchases C_i .*

Proof. Solve by backward induction.

Stage 4 (purchase, given menu and disclosure): A tested type- i individual faces the choice $\{C_i\}$ (since C_j , $j \neq i$, is unavailable to her under the conditioning). She purchases C_i . An untested individual faces $\{C_U\}$ and purchases C_U (since the alternative, no insurance, yields strictly less by Proposition 1).

Stage 3 (test decision): By Lemma 1, testing strictly dominates not testing for the individual. Hence every individual tests.

Stage 2 (insurer offers): Given that all individuals test, the insurer’s posted menu is best-responding: each contract is actuarially fair given the conditioning, so expected profit is zero on each, and no deviation strictly increases profits while still attracting customers (the contracts are at the customers’ bliss points within the actuarial-fairness constraint).

Off-path beliefs: If an individual deviates by not testing, the insurer offers C_U , which is sequentially rational since beliefs over untested individuals are simply the prior (θ_H, θ_L) .

All conditions of subgame perfection are met. $\square$

3.2 Code of Conduct

Here insurers observe *whether* an individual has tested but not the *result*.

Proposition 3 (Code of conduct equilibrium). *There is a weak perfect Bayesian equilibrium in which*

1. *the menu offered to tested individuals is $\{C_H, C_L\}$ and to untested individuals is $\{C_U\}$;*
2. *every individual tests;*
3. *tested individuals truthfully self-select into C_i .*

Proof. The only subtlety relative to disclosure duty is that, post-testing, the insurer cannot directly observe type, so the menu $\{C_H, C_L\}$ must be incentive-compatible.

IC check. A tested type- H individual’s payoff from C_H is $u(w - \mu H)$. From C_L it is

$$\mathbb{E}_\mu[u(w - \mu L - H + L), u(w - \mu L)] = \mu u(w - \mu L - (H - L)) + (1 - \mu)u(w - \mu L).$$

The mean wealth under C_L for type H is $w - \mu L - \mu(H - L) = w - \mu H$, the same as C_H . Since C_H delivers this with certainty and C_L delivers a mean-preserving spread of it, strict concavity gives $u(w - \mu H) > \text{EU}(C_L \mid H)$. Symmetric argument for type L . So both IC constraints are slack.

Beliefs. On the equilibrium path, the insurer’s belief at “tested and chose C_i ” is “type i with probability 1.” For untested individuals, beliefs are the prior. Off-path beliefs (e.g., a tested individual chooses a contract outside the menu) can be set to assign probability 1 to type H , which deters deviations.

Testing decision and insurer offers follow as in the disclosure-duty case. $\square$

3.3 Consent Law

The insurer cannot observe test status. The individual may voluntarily reveal a test result; concealment is permitted.

Proposition 4 (Consent law equilibrium). *A Bayesian Nash equilibrium exists in which insurers offer $\{C_H, C_L\}$, with C_L conditional on the individual presenting a verified test result certifying type L ; every individual tests; each type purchases the appropriate contract.*

Proof. The key observation is that *the absence of the principle of indemnity removes any incentive for the insurer to fear adverse selection at the pooled contract.*

Suppose for contradiction that the insurer also offered $C_U = (\mu, d_{U,np}^*)$ available to everyone. Then a tested type- L individual would prefer C_L to C_U (Lemma 1 applied at the individual level), and would reveal her test result. A tested type- H individual, if she could also access C_L , would prefer it—but C_L is conditional on verified type- L test results, which a type- H cannot produce.

So her choice is between C_H and C_U , and by Lemma 1 she prefers C_H . Hence C_U attracts no one, and the equilibrium is unchanged whether it is offered or not.

Now check no profitable deviation exists. Any deviation contract must satisfy actuarial fairness in expectation given the buyers it attracts. The equilibrium contracts C_H and C_L each give the respective type her bliss point at zero insurer profit. No alternative contract can give a type strictly higher utility while remaining actuarially fair, by the standard Mossin (1968) result.

Testing: By Lemma 1 and the conditioning of C_L on verified L -test results, every individual strictly prefers to test (untested, she would only be able to access C_H , which is dominated for type L). $\square$

3.4 Information Ban

Insurers cannot condition on test status or results; individuals cannot reveal.

Proposition 5 (Information ban equilibria). *Under the information ban, there exist two perfect Bayesian equilibria:*

1. **Separating equilibrium:** Insurers offer $\{C_H, C_L\}$ (without conditioning); every individual tests; each type self-selects into the appropriate contract.
2. **Pooling equilibrium:** Insurers offer only $C_U = (\mu, d_{U,np}^*)$; no individual tests; everyone purchases C_U .

Proof. Separating equilibrium: The IC check is identical to the code-of-conduct case (the same argument that type H strictly prefers C_H to C_L goes through). The insurer cannot exclude any individual from any contract—but this doesn't matter, because incentive compatibility already does the work. By Lemma 1, individuals test.

Pooling equilibrium: If only C_U is offered, no individual tests (Lemma 1 doesn't bite because the alternative to C_U is no insurance, not a type-specific contract). The insurer's prior remains (θ_H, θ_L) ; off-path beliefs after any deviation contract can be set to make deviations unprofitable. $\square$

Remark. This is an important correction to the original paper. The original statement that the separating outcome is “the” perfect Bayesian equilibrium under the information ban is incorrect without an equilibrium-selection refinement (e.g., the intuitive criterion or Miyazaki-Spence cross-subsidy refinement).

4 Proposition 4: No Testing Under Principle of Indemnity

Lemma 2. *With the principle of indemnity, the uninformed individual's optimal indemnity is $d_{U,p}^* = H$, and the actuarially fair premium rate is $q(H) = \mu(\theta_H + \theta_L L/H)$.*

Proof. Let $\Psi(d) := \sum_i \theta_i \mu u(w - q(d)d - i + \min\{i, d\}) + (1 - \mu)u(w - q(d)d)$ where $q(d)$ is set to make the contract actuarially fair: $q(d)d = \mu[\theta_H \min\{H, d\} + \theta_L \min\{L, d\}]$.

We analyze three regions:

Region 1: $d \in [0, L]$. Here $\min\{i, d\} = d$ for both types, so $q(d) = \mu$. Loss-state wealth is $w - \mu d - i + d = w - \mu d - i + d$. Differentiating Ψ ,

$$\Psi'(d) = \mu(1 - \mu) \left[\sum_i \theta_i u'(w - \mu d - i + d) - u'(w - \mu d) \right].$$

At $d = L$, by the same Jensen argument as in Proposition 1, $\Psi'(L^-) > 0$. So the optimum is not in $[0, L]$.

Region 2: $d \in [L, H]$. Now $\min\{L, d\} = L$ and $\min\{H, d\} = d$, so $q(d)d = \mu[\theta_H d + \theta_L L]$, hence $q(d) = \mu[\theta_H + \theta_L L/d]$ and the premium is $\pi(d) = \mu(\theta_H d + \theta_L L)$.

The type- L loss-state wealth is $w - \pi(d) - L + L = w - \pi(d) = W_0$. So in this region, type L has full coverage. Type H 's loss-state wealth is $w - \pi(d) - H + d$.

The objective becomes

$$\Psi(d) = \theta_L \mu u(w - \pi(d)) + \theta_H \mu u(w - \pi(d) - H + d) + (1 - \mu)u(w - \pi(d)).$$

Differentiating with respect to d , using $\pi'(d) = \mu\theta_H$:

$$\begin{aligned}\Psi'(d) &= -\mu\theta_H(\theta_L\mu + 1 - \mu)u'(w - \pi(d)) + \theta_H\mu(1 - \mu\theta_H)u'(w - \pi(d) - H + d) \\ &= \mu\theta_H(\theta_L\mu + 1 - \mu)[u'(w - \pi(d) - H + d) - u'(w - \pi(d))].\end{aligned}$$

For $d < H$, the argument $w - \pi(d) - H + d < w - \pi(d)$, so by $u'' < 0$, u' at the smaller argument is larger, hence $\Psi'(d) > 0$ on $[L, H]$.

Region 3: $d > H$. Now $\min\{i, d\} = i$ for both types, the indemnity is capped at the loss, and increasing d only increases the premium without changing payouts. So $\Psi'(d) < 0$ for $d > H$.

Combining: Ψ is increasing on $[0, H]$ and decreasing on $[H, \infty)$, so $d_{U,p}^* = H$. $\square$

Proposition 6 (No testing equilibrium with principle of indemnity). *Under the principle of indemnity, in the disclosure-duty and code-of-conduct regimes, the unique equilibrium has*

1. *insurers offering $\{C_H, C_L, C'_U\}$ where $C'_U = (q(H), H)$ and the conditioning is as before;*
2. *no individual tests;*
3. *every individual purchases C'_U .*

Proof. It suffices to show that, given the menu, *no individual prefers to test.*

Key observation. Under C'_U , the payout in the loss state is $\min\{i, H\} = i$ for both types. So loss-state wealth under C'_U is $w - q(H)H - i + i = w - q(H)H$, and no-loss wealth is also $w - q(H)H$. Wealth is constant across all states.

Compare to the expected utility from testing. After testing and purchasing C_i , type- i 's wealth is $w - \mu i$ with certainty. Ex ante (before knowing one's type), the expected utility from testing is

$$V_{\text{test}} = \theta_H u(w - \mu H) + \theta_L u(w - \mu L).$$

The expected utility from C'_U is

$$V_{\text{pool}} = u(w - q(H)H).$$

Recall $q(H)H = \mu(\theta_H H + \theta_L L) = \mu A$. So $w - q(H)H = w - \mu A = \theta_H(w - \mu H) + \theta_L(w - \mu L)$.

Hence

$$V_{\text{pool}} = u(\theta_H(w - \mu H) + \theta_L(w - \mu L)) > \theta_H u(w - \mu H) + \theta_L u(w - \mu L) = V_{\text{test}}$$

by strict concavity of u (Jensen's inequality for concave functions, using $u'' < 0$, not $u''' > 0$).

So no individual tests. The insurer's offering of C'_U is actuarially fair given that everyone purchases it. The other contracts C_H, C_L are offered but unused.

Uniqueness: Any equilibrium must have all individuals not testing (by the above), and any actuarially fair pooled contract offered to everyone must, by Lemma 2, have $d = H$. So C'_U is the unique pooled contract. $\square$

Remark. Correction. The original proof says “individuals who are prudent will prefer certainty over uncertainty.” This is incorrect: it is *risk aversion* ($u'' < 0$, concavity), not *prudence* ($u''' > 0$, convexity of u'), that delivers the result. Prudence is needed for Proposition 1 (where the inequality direction goes the *other* way), not Proposition 4. This is a mathematical error in the original paper.

5 Summary of Key Corrections

1. **Proposition 1:** Added explicit existence/uniqueness argument via strict concavity. Provided clean Jensen-based derivation of $d^* > A$. Clarified role of prudence assumption.
2. **Proposition 2:** Separated the four regime arguments with proper equilibrium concept treatment (game form, strategies, beliefs, SPE/PBE verification). **Corrected:** The information-ban case admits *multiple* PBEs; the original statement is incorrect without an equilibrium-selection refinement.
3. **Proposition 4: Corrected conceptual error**—the proof invokes prudence where it should invoke risk aversion. The corrected proof uses Jensen on u concave, not u' convex. Added Lemma 2 with full piecewise analysis of the corner solution $d_{U,p}^* = H$.
4. **Propositions 6 and 10:** The IC analysis in the original paper is incomplete. All pairwise IC constraints should be listed and the binding ones identified before claiming a particular menu is the equilibrium.
5. **Value-of-information appendix:** The inequalities should be proved as propositions with conditions for strictness, not left as concluding remarks.

Most Critical Issues

- The Proposition 4 error (prudence vs. risk aversion) is the most important to fix before submission—it’s the kind of error that gets a paper desk-rejected if a careful referee spots it.
- The Proposition 2 multiplicity issue under the information ban is second-most important, because it affects the policy conclusion (the paper claims separating outcomes “arise” under the information ban, but they only arise under a particular equilibrium selection).